1) Given $\vec{u} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$, $\vec{v} = \begin{bmatrix} 4 \\ 1 \end{bmatrix}$, find $\text{proj}_u(v)$.

$$\text{proj}_u(v) = \frac{v \cdot u}{u \cdot u} u = \frac{4 \cdot 1 + 1 \cdot 2}{1 \cdot 1 + 2 \cdot 2} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \frac{6}{3} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 2 \\ 4 \end{bmatrix}$$

2) Using the result from ①, compute the error vector

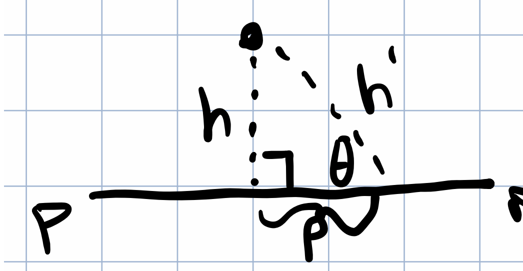
$$\vec{e} = \vec{v} - \text{proj}_u(\vec{v})$$

and verify that $\vec{e} \cdot \vec{u} = 0$

$$\vec{e} = \begin{bmatrix} 4 \\ 1 \end{bmatrix} - \begin{bmatrix} 2 \\ 4 \end{bmatrix} = \begin{bmatrix} 2 \\ -3 \end{bmatrix}$$

$$\vec{e} \cdot \vec{u} = \begin{bmatrix} 2 \\ -3 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 2 \end{bmatrix} = 2 \cdot 2 + (-3 \cdot 1) = 4 - 4 = 0 \checkmark$$

3) Explain why the shortest path from a point to a plane is a perpendicular line.



Any h' with an angle less than $\pi/2$ is longer than h as they form a right triangle

plane projecting into/out of the plane of the page

Proof:

Suppose $h \leq h' \rightarrow h' = h + a$

" $\theta \le \pi/2$ $\theta = \pi/2 - \phi$

$h = h' - a$

$\text{proj}_{h'}(P) = \rho$ $\rho^2 + h^2 = h'^2$

S	C	T
O	A	O
H	H	A

$\tan(\theta) = \frac{h}{\rho} \rightarrow \rho \tan(\theta) = h$

$\rho^2 + h^2 = h'^2$

$\cos\theta = \frac{h'-a}{h'}$ $h'\cos\theta = h$

$\rho^2 + (h'-a)^2 = h'^2$

$\sin\theta = \frac{\rho}{h'}$ $h'\sin\theta = \rho$

$\rho^2 + h'^2 - 2ah' - a^2 = h'^2$

$\rho > 0$ $a > 0$
$h > 0$

$\rho^2 - 2ah' + a^2 = 0$

$h'^2 \sin^2\theta - 2ah'\cos\theta + a^2 = 0$

$h'^2(1 - \cos^2\theta) - 2ah'\cos\theta + a^2 = 0$

$h'^2 - h'^2\cos^2\theta - 2ah'\cos\theta + a^2 = 0$

$h'^2\cos^2\theta + 2ah'\cos\theta - a^2 - h'^2 = 0$

$\frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ $x = \cos\theta$

$h'^2 x^2 + 2ah'x - a^2 - h'^2 = 0$

$\cos\theta = -2ah' \pm \sqrt{4a^2h'^2 - 4h'^2(-a^2 - h'^2)}$

$$\frac{\qquad}{2h'^2}$$

$$= \frac{-\cancel{2}a\cancel{h'} \pm \cancel{2}\cancel{h'}\sqrt{a^2+a^2-h'^2}}{\cancel{2}h'^{\cancel{2}}}$$

$$\cos\theta = \frac{-a \pm \sqrt{2a^2-h'^2}}{h'}$$

$$\frac{h' \cancel{-a}}{\cancel{h'}} = \frac{\cancel{-a} \pm \sqrt{2a^2-h'^2}}{\cancel{h'}}$$

$$h' = \pm\sqrt{2a^2-h'^2}$$

$$h'^2 = 2a^2-h'^2$$

$$0 = 2(a^2-h'^2)$$

$$0 = (a+h')(a-h')$$

$$h' = \pm a \qquad h' = h+a$$

$$a+h' = 0 \qquad a-h' = 0$$

$$h' = -a \qquad h' = a$$

$$\cancel{h' = h+a} \qquad h+a = a$$

$$h = 0 \;\therefore\; h' = 0$$

$\cancel{h = h' + a}$

$\cancel{a = 0}$

~~contradiction~~

$h = h'$

point on the plane

$\therefore h < h'$, so the perpendiculal line connecting a point to a plane is the shortest path

4) Suppose you are fitting a line to data, but no point passes through every point. Why is minimizing the squared error a reasonable goal instead of trying to eliminate every error.

We are already supposing the data can not fit on a single line, so there will always be error, and trying will be unreasonable.

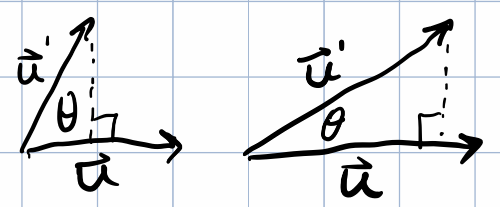
Instead, we can project to the closest point on the line that minimizes the squared error, which gives us the best fit to the data

5) Suppose $\vec{u} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$, without doing much arythmatic,

What is $\text{proj}_{\vec{u}}(\vec{u})$? Why?

$\text{proj}_{\vec{u}}(\vec{u}) = \vec{u}$ because the projection is the component of a vector that matches the direction of the vector that it is being projected onto, and when those vectors match, the component of the projecting vector is the entire vector.

$|\vec{u}| = |\vec{u}'|$



as θ approaches 0, $\vec{u}'$ approaches $\vec{u}$

also $\mathrm{proj}_{\vec{u}}(\vec{u}) = \frac{\cancel{\vec{u}\cdot\vec{u}}}{\cancel{\vec{u}\cdot\vec{u}}}\vec{u}$